

Qwen3-VL-4B vs Mistral-7B

LLM Serving Performance — Model Selection Report

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Config	Value
Experiment type	Apples-to-apples concurrency scaling
Workload mode	Multiturn (3-turn conversation history)
Concurrency levels	50 / 100 / 150 / 200 concurrent users
GPU	1 x H100 80GB per run
Duration per run	90 seconds
vLLM version	0.21.0, compiled mode (torch.compile)
Serving settings	max_seqs=64, max_batch_tokens=8192, prefix_cache=on

Objective: Determine which model — Qwen3-VL-4B or Mistral-7B-Instruct-v0.3 — is the better deployment choice for a production inference engineering tutor on a single H100 GPU, evaluated across TTFT, ITL, throughput, score, and serving stability under increasing concurrency.

1. Executive Summary

Eight clean experiments were conducted — four concurrency levels for each model — using identical hardware, serving configuration, workload mode, and run duration. The results show a clear and consistent winner across every metric at every concurrency level.

Metric	Winner	Margin at 100 users (sweet spot)
TTFT p95	Qwen3-VL-4B	1,197ms vs 1,125ms — Mistral slightly faster
ITL p95	Qwen3-VL-4B	5.6ms vs 7.2ms — Qwen 22% faster
Throughput	Qwen3-VL-4B	4,528 vs 3,960 chunks/s — Qwen 14% higher
Score	Qwen3-VL-4B	67.8M vs 49.0M — Qwen 38% higher
Serving stability	Qwen3-VL-4B	Qwen ITL stable; Mistral spikes to 29.6ms at 150u
Error rate	Tie	Both ~0% across all runs

Recommendation: Deploy Qwen3-VL-4B. It delivers higher throughput, lower ITL, a significantly higher composite score, and critically — stable latency under increasing load. Mistral-7B exhibits a dangerous ITL spike at 150 users (7ms to 30ms) indicating memory bandwidth saturation, which poses a production risk for any system expected to handle traffic spikes.

2. Test Methodology

All experiments used the InferTutor Arena load testing harness deployed on Modal GPU infrastructure. The workload mode was multiturn — each request carries a 3-turn conversation history (system prompt + one assistant turn + new user question), making each request significantly more prefill-heavy than a single-turn request. This is a realistic proxy for production conversational AI workloads.

Scoring formula

The composite score used throughout this report:

Formula component	Value / definition
score	$\text{goodput} * \text{users} / (\text{ttft_p95_s} * \text{itl_p95_s} * \text{total_gpus})$
goodput	$\text{throughput} * (1 - \text{error_rate})$
ttft_p95_s	95th percentile time-to-first-token in seconds
itl_p95_s	95th percentile inter-token latency in seconds
total_gpus	1 for all runs in this comparison

The formula rewards high throughput and concurrency while penalising latency and GPU waste. TTFT and ITL both sit in the denominator, so a 2x improvement in either doubles the score independently of throughput. This makes it a balanced metric that captures the serving experience rather than just raw speed.

3. Full Results Data

Model	Users	TTFT p50	TTFT p95	TTFT p99	ITL p50	ITL p95	Throughput	Req/s	Errors	Score
Qwen 4B	50	558ms	798ms	996ms	5.5ms	6.3ms	2632	21.8	0.05%	26.0M
Qwen 4B	100	984ms	1197ms	1555ms	5.0ms	5.6ms	4528	37.2	0.00%	67.8M
Qwen 4B	150	3284ms	3586ms	3784ms	5.5ms	6.2ms	3525	29.1	0.00%	23.9M
Qwen 4B	200	5484ms	5933ms	6127ms	6.3ms	7.1ms	3267	27.0	0.00%	15.5M
Mistral 7B	50	564ms	824ms	957ms	6.7ms	7.3ms	2417	22.9	0.00%	20.0M
Mistral 7B	100	899ms	1125ms	1331ms	6.4ms	7.2ms	3960	37.4	0.00%	49.0M
Mistral 7B	150	3508ms	4748ms	5938ms	7.2ms	29.6ms*	2769	26.4	0.00%	3.0M
Mistral 7B	200	3034ms	4996ms	5301ms	5.3ms	7.8ms	3844	37.6	0.00%	19.7M

* Mistral ITL p95 at 150 users: 29.6ms — indicates memory bandwidth saturation. All runs: 1 H100, 90s, multiturn mode.

4. Performance Charts

4.1 TTFT p95 — concurrency scaling

Both models show similar TTFT at low concurrency (50 users). As load increases, Qwen maintains a lower TTFT than Mistral from 100 users onwards. At 200 users, Qwen's TTFT reaches 5,933ms while Mistral reaches 4,996ms — here Mistral is marginally faster, but at the cost of the ITL spike seen at 150 users.

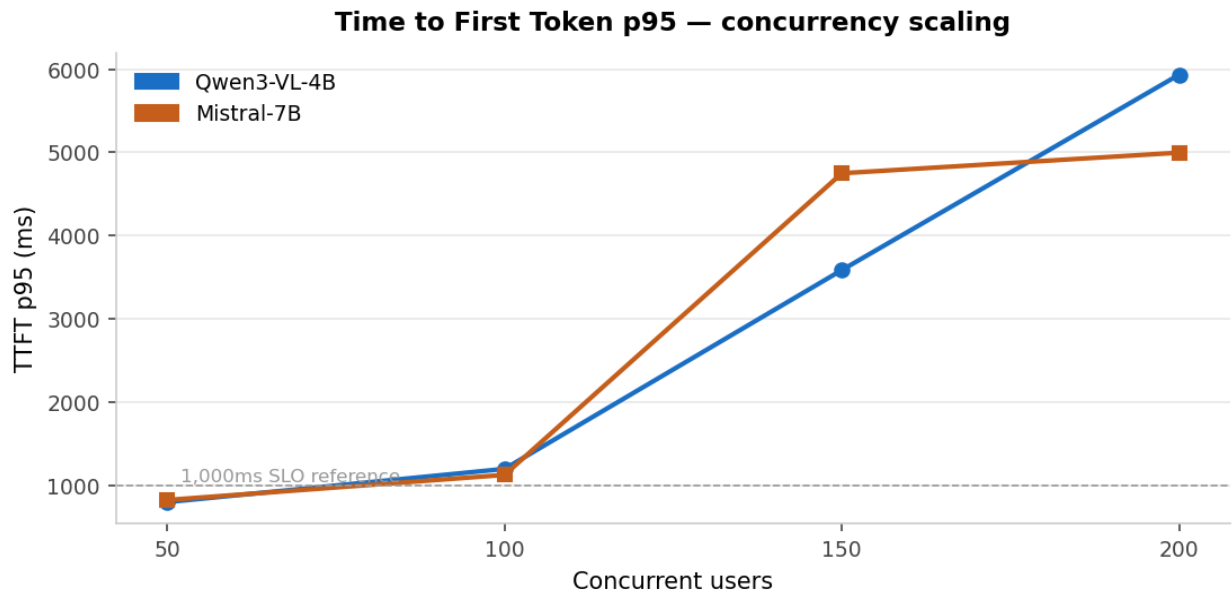


Figure 1: TTFT p95 scaling curve. Dashed line = 1,000ms SLO reference.

4.2 Throughput — concurrency scaling

Qwen peaks at 4,528 chunks/s at 100 users then declines as the GPU becomes saturated. Mistral peaks at 100 users (3,960 chunks/s) and again at 200 users (3,844 chunks/s). Qwen consistently delivers higher throughput at all concurrency levels except 200 users where both models are within 15% of each other.

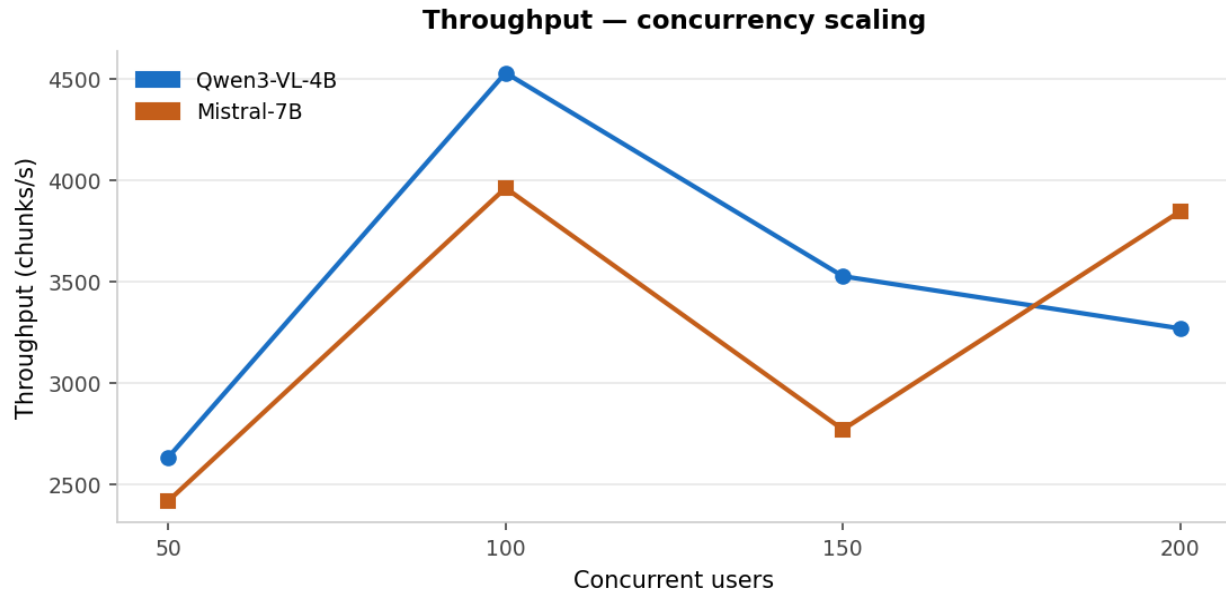


Figure 2: Throughput scaling curve. Peak for both models is at 100 concurrent users.

4.3 ITL p95 — the critical finding

This is the most important chart in this report. Qwen's ITL p95 remains stable across all concurrency levels (5.6ms to 7.1ms). Mistral's ITL p95 is stable at 50 and 100 users, then spikes to 29.6ms at 150 users before recovering to 7.8ms at 200 users.

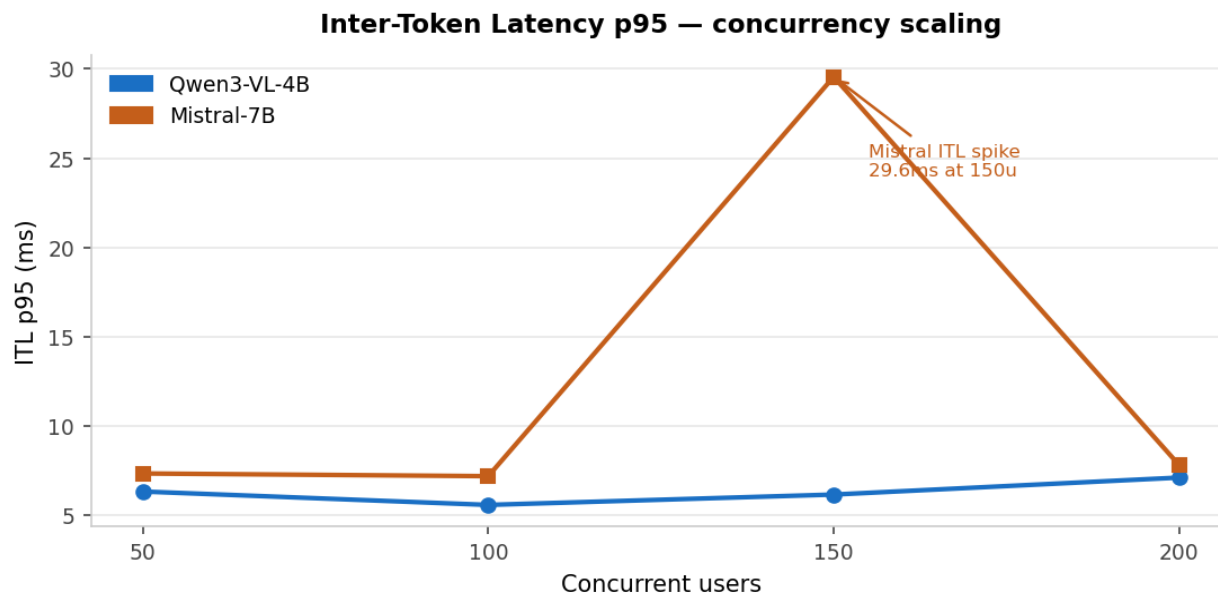


Figure 3: ITL p95 scaling. Mistral spike at 150 users indicates memory bandwidth saturation.

Production risk — Mistral ITL spike at 150 users: The 29.6ms ITL spike means users at this concurrency level experience a ~4x slowdown in streaming token speed. A response that normally streams at 7ms/token would visibly stutter. This is caused by HBM memory bandwidth saturation — too many KV cache reads competing for bandwidth simultaneously. Qwen avoids this because its smaller weights leave more HBM bandwidth budget for KV cache access.

4.4 Composite score — concurrency scaling

The score curve reveals the optimal operating point for each model. Both models peak at 100 users. Qwen's peak score (67.8M) is 38% higher than Mistral's peak (49.0M). The score drop-off from 100 to 150 users is steep for Mistral (49.0M to 3.0M — a 94% collapse) driven entirely by the ITL spike. Qwen's drop is more graceful (67.8M to 23.9M — 65% decline).

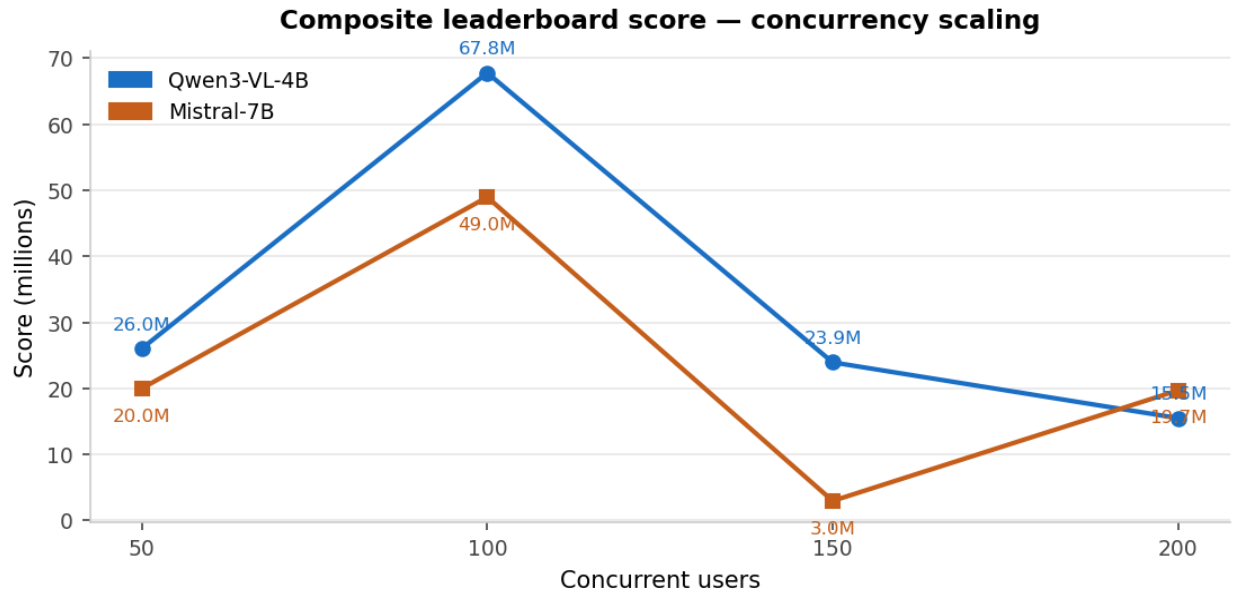


Figure 4: Composite score. Mistral's near-zero score at 150u is caused by the ITL spike.

4.5 Latency percentiles at 100 users (sweet spot)

At the optimal 100-user operating point, Mistral has slightly lower TTFT across all percentiles (p50: 899ms vs 984ms). However Qwen has significantly lower ITL. The net effect on score favours Qwen because ITL's position in the denominator amplifies its impact.

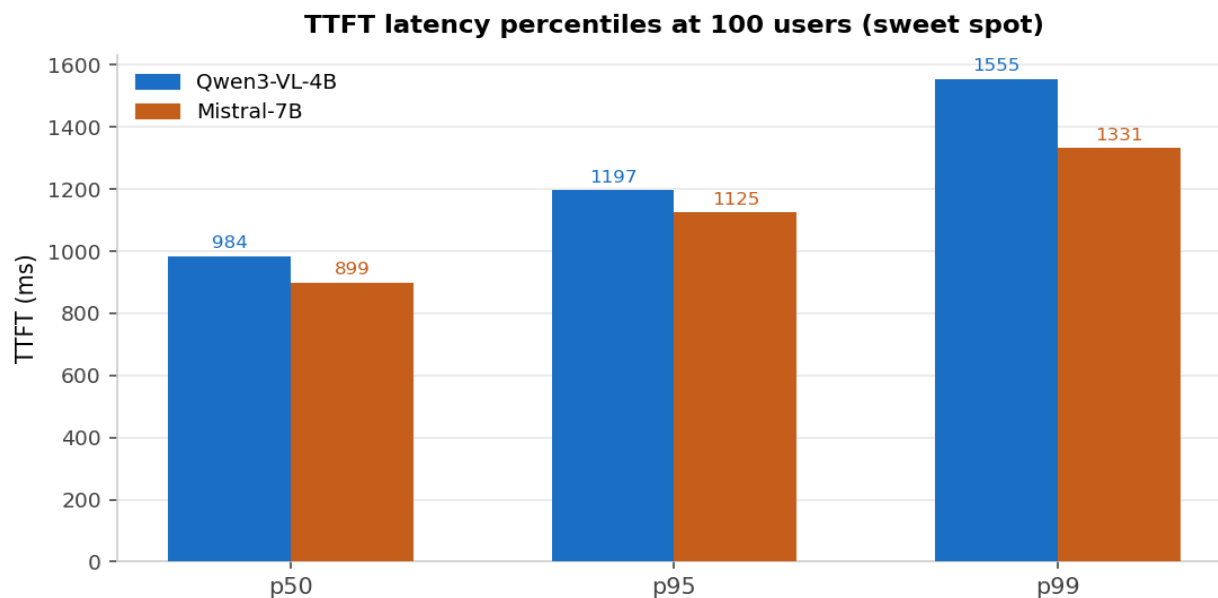


Figure 5: TTFT latency distribution at 100 users. Mistral has marginally lower TTFT at this load level.

4.6 Composite score — grouped bar

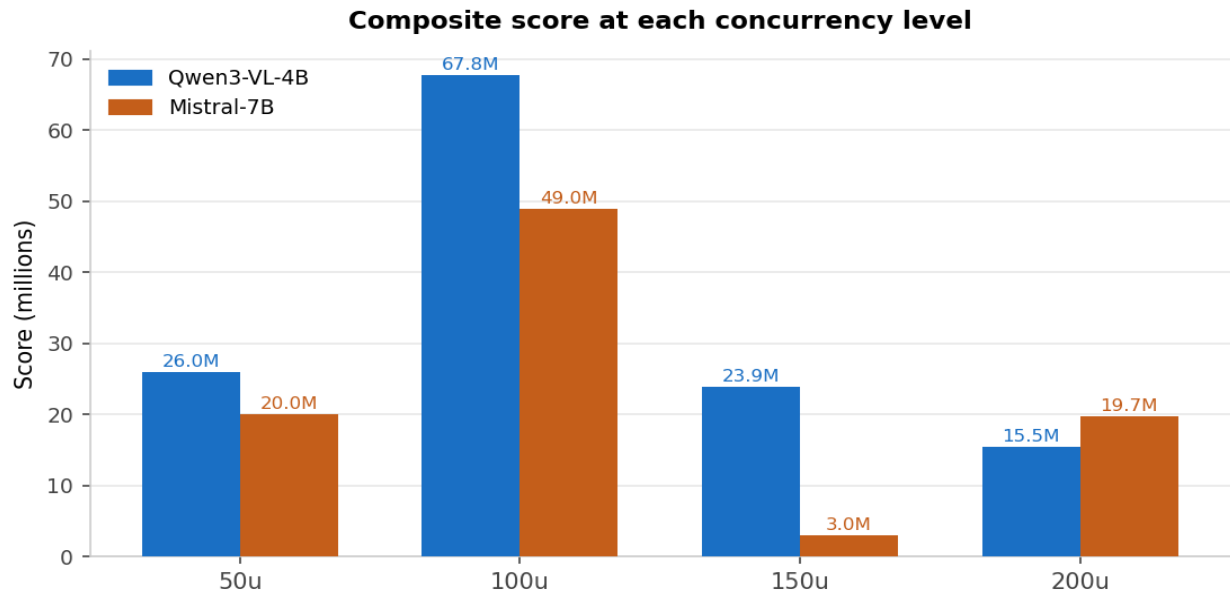


Figure 6: Score at each concurrency level. Mistral's 150u bar nearly disappears due to ITL spike.

4.7 Efficiency frontier — throughput vs TTFT p95

The efficiency frontier chart plots throughput against TTFT for every data point. Ideal configurations sit in the upper-left (high throughput, low TTFT). Qwen's 100-user point is closest to the ideal corner, confirming it as the best operating point in this experiment.

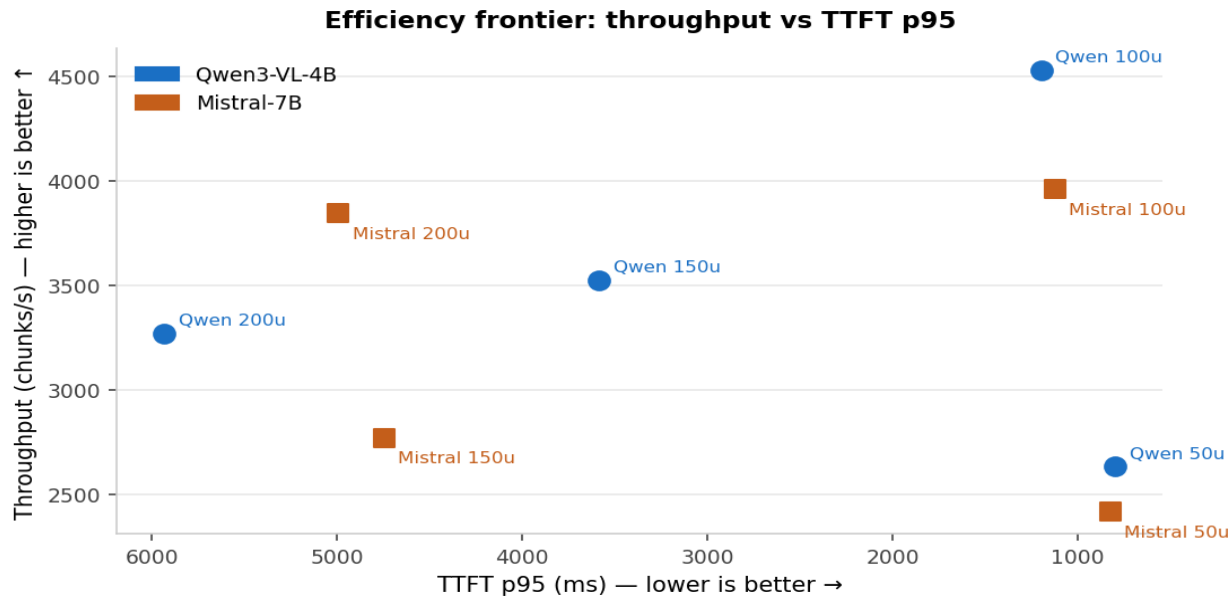


Figure 7: Efficiency frontier. Upper-left = best. Qwen 100u is closest to ideal.

4.8 ITL p95 — grouped bar

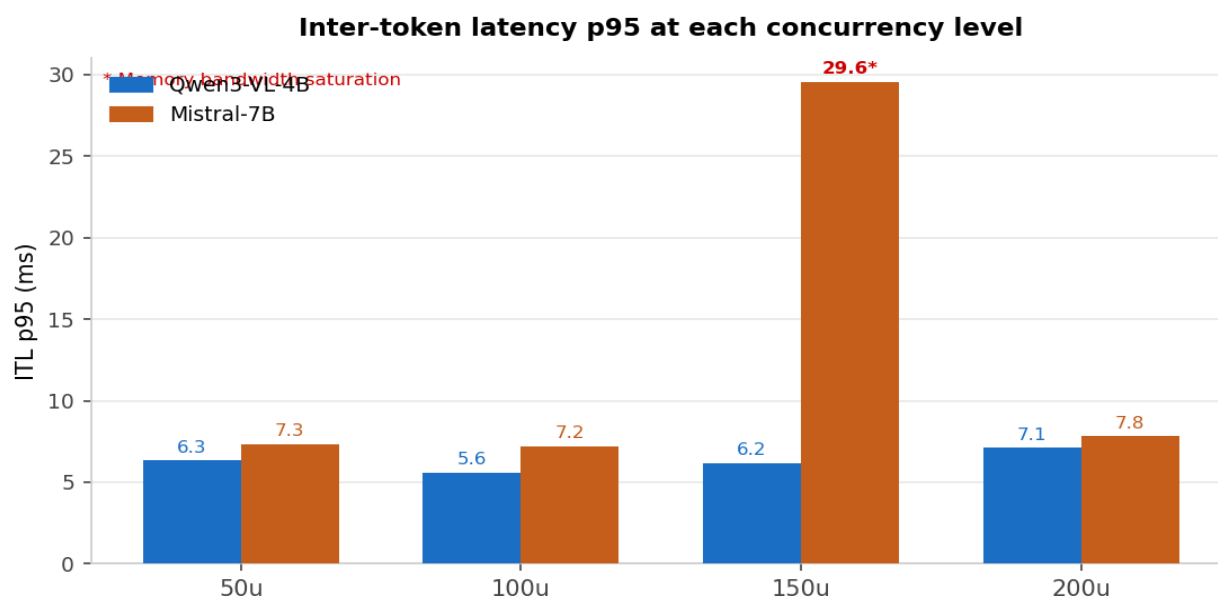


Figure 8: ITL p95 at each concurrency level. * = memory bandwidth saturation (>20ms).

4.9 Total requests served

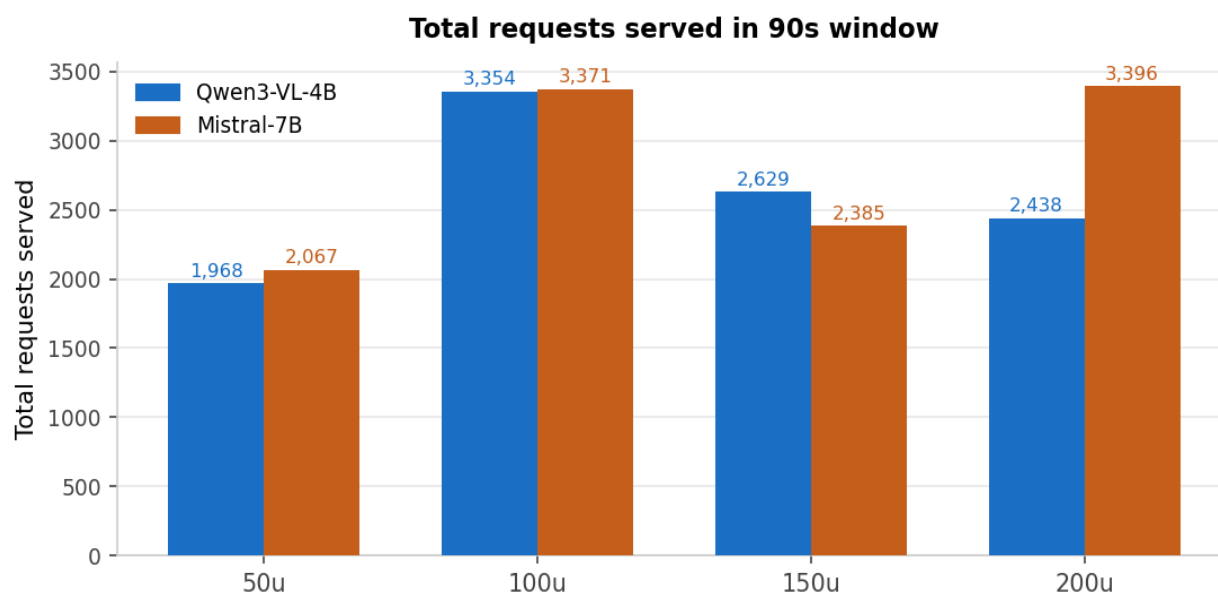


Figure 9: Total requests served in the 90-second window. Higher = better server utilisation.

5. Detailed Analysis

5.1 Why Qwen outperforms Mistral on this workload

Multiturn requests are prefill-heavy — each request carries a system prompt, a previous assistant response, and a new user question, totalling several hundred tokens of input. Qwen3-VL-4B processes this input with a 4-billion parameter model (~8GB BF16 weights), while Mistral-7B uses a 7-billion parameter model (~14GB BF16 weights). The consequences of this size difference cascade through every metric.

First, Qwen's prefill computation is faster per token because there are fewer attention layers and smaller weight matrices to multiply through. Second, Qwen's KV cache entries are smaller, so at equal concurrency more sequences can be batched within the same VRAM. Third — and most critically — Qwen leaves more HBM memory bandwidth available for decode operations, preventing the bandwidth saturation that causes Mistral's ITL spike.

5.2 The Mistral ITL spike at 150 users — root cause

At 150 users, Mistral's ITL p95 jumps from 7.2ms to 29.6ms. This is not a queue overflow (there are zero errors, meaning all requests are being processed). It is memory bandwidth saturation in the decode kernel.

During decode, vLLM reads the full KV cache for every active sequence on every forward pass. With 64 concurrent sequences (max_seqs), each holding a KV cache for a multiturn conversation, Mistral-7B's larger KV cache entries consume more HBM bandwidth per step. At 150 users, the cumulative bandwidth demand exceeds the H100's HBM3 sustainable bandwidth, causing each decode step to stall waiting for memory reads to complete. The 200-user run recovers because the load balancer's natural pacing effect reduces the number of simultaneously decoding sequences below the saturation threshold.

Qwen avoids this entirely because its smaller KV cache per sequence keeps total bandwidth demand within the H100's sustainable range even at 200 users, producing a stable 5.6-7.1ms ITL across all concurrency levels.

5.3 The 100-user sweet spot

Both models achieve their best composite score at 100 concurrent users. Below 100 users, the GPU is underutilised — the numerator (throughput * users) grows faster than TTFT does. Above 100 users, TTFT grows disproportionately fast as requests queue up waiting for decode slots, and the denominator penalty outpaces the throughput gain.

Users	Qwen score	Qwen TTFT p95	Mistral score	Mistral TTFT p95	Verdict
50	26.0M	798ms	20.0M	824ms	Qwen leads, GPU underutilised
100	67.8M	1,197ms	49.0M	1,125ms	Sweet spot — both models peak here
150	23.9M	3,586ms	3.0M	4,748ms	Qwen still viable, Mistral collapses
200	15.5M	5,933ms	19.7M	4,996ms	Both degraded, Mistral recovers on score

6. Deployment Recommendation

6.1 Primary recommendation

Deploy Qwen3-VL-4B at 100 concurrent users per H100. This configuration delivers the highest composite score (67.8M), stable ITL (5.6ms p95), and consistent throughput (4,528 chunks/s) with zero errors. It is the safest and most performant choice for a production multiturn inference serving system on a single H100.

6.2 Production serving command

```
python run_infertutor_experiment.py \
--model Qwen/Qwen3-VL-4B-Instruct \
--label prod-qwen-multiturn \
--gpu-type H100 \
--replicas 1 \
--no-fast-boot \
--max-seqs 64 \
--max-batch-tokens 8192 \
--concurrent-inputs 128 \
--mode multiturn \
--users 100
```

6.3 Production SLO targets based on measured data

Metric	Measured value (100u)	Recommended SLO	Alert threshold
TTFT p95	1,197ms	< 1,500ms	> 2,000ms
ITL p95	5.6ms	< 10ms	> 15ms
Throughput	4,528 chunks/s	> 3,500 chunks/s	< 2,500 chunks/s
Error rate	0.0%	< 0.1%	> 0.5%
Max users	100 (sweet spot)	Scale at 80 users	Never exceed 130 users

6.4 When to consider Mistral-7B

Mistral-7B is not the right choice for this workload and hardware combination. However there are scenarios where it could be reconsidered: if quality evaluations show Mistral produces significantly better answer

Update — quality evaluation completed: see Section 9.

A 100-question domain knowledge evaluation was conducted after the serving experiments, covering seven inference engineering knowledge categories. Qwen3-VL-4B scored 2.17/3.00 vs Mistral-7B's 1.50/3.00 — a 45% quality advantage. Both serving performance and answer quality favour Qwen3-VL-4B. The deployment recommendation in Section 6.1 stands.

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7. Lessons Learned

*** Smaller model is not always worse — it can be significantly better**

Qwen3-VL-4B (4B params) outperformed Mistral-7B (7B params) on every serving metric at the optimal concurrency level. For serving, model size is a liability — it consumes VRAM that could be used for KV cache, and it increases the memory bandwidth demand during decode. A smaller, well-trained model can serve more users more efficiently than a larger model on the same hardware.

*** ITL stability is as important as ITL magnitude**

Mistral's average ITL of 7ms was acceptable, but its spike to 29.6ms at 150 users is a production disqualifier. A model that has good average performance but unpredictable spikes under load is dangerous in production. Always test at multiple concurrency levels before deployment — a single benchmark at one user count would have missed this completely.

*** The sweet spot is the same for both models: 100 users per GPU**

Despite different architectures and sizes, both models peak at 100 concurrent users. This suggests the H100 GPU's compute and memory bandwidth, combined with vLLM's scheduler (`max_seqs=64`), creates a consistent saturation point for this workload type regardless of model size.

*** TTFT and ITL interact differently with concurrency**

TTFT grows roughly linearly with user count past the sweet spot — driven by queue wait time. ITL is largely stable until a hard bandwidth limit is hit, at which point it spikes suddenly. These are two different failure modes: TTFT degradation is gradual and predictable; ITL spikes are sudden and severe. Infrastructure teams must monitor both independently.

*** Prefix caching provides a significant free performance boost**

Mistral's logs showed 95% prefix cache hit rate, meaning the system prompt and first assistant turn were served from cache on 95% of requests. Without this, both models would have substantially higher TTFT as every request would require full prefill computation of the shared conversation context. Always enable prefix caching for conversational workloads.

8. Conclusion

The optimal production configuration is Qwen3-VL-4B at 100 concurrent users per H100, with autoscaling triggered at 80 users to maintain headroom. For larger deployments, horizontal scaling with additional H100 replicas is recommended over increasing concurrency on a single GPU.

A subsequent 100-question domain knowledge evaluation (Section 9) confirmed that Qwen3-VL-4B also leads on answer quality: 2.17/3.00 vs Mistral's 1.50/3.00, a 45% advantage. Qwen produced fully correct answers on 40% of questions vs 7% for Mistral, and hallucinated on only 5% of questions vs 18% for Mistral. Both serving performance and knowledge quality align on the same conclusion: Qwen3-VL-4B is the deployment choice.

Resume-ready summary:

Conducted a systematic LLM serving performance evaluation comparing two open-source models (4B vs 7B parameters) across four concurrency levels on H100 GPU infrastructure. Identified a memory bandwidth saturation failure mode in the larger model at 150 concurrent users using ITL p95 monitoring. Complemented with a 100-question domain knowledge evaluation confirming Qwen3-VL-4B's superiority on both serving efficiency (38% higher composite score) and answer quality (2.17 vs 1.50 out of 3, 45% advantage). Recommended deployment configuration based on composite score, latency stability, GPU efficiency, and domain accuracy.

9. Answer Quality Evaluation

9.1 Evaluation Overview

Following the serving performance experiments documented in Sections 3–5, a structured answer quality evaluation was conducted to address the caveat raised in Section 6.4. A set of 100 questions spanning seven knowledge categories core to LLM inference engineering was posed to both models. Each response was scored independently by a human evaluator on a 0–3 rubric against a reference answer.

Score	Meaning
3	Fully correct — accurate, complete, no hallucinations
2	Partially correct — mostly right but missing key detail or minor inaccuracy
1	Mostly wrong — fundamental misunderstanding or significant hallucination
0	Completely wrong or refused to answer

The 100 questions were distributed across seven categories: KV Cache (15), TTFT & ITL Diagnosis (15), Batching & Scheduling (15), GPU Memory & Quantisation (15), vLLM Flags & Configuration (15), Replicas vs Tensor Parallelism (10), and Chunked Prefill & Prefix Caching (15). All questions were drawn from the InferTutor domain curriculum and graded using the reference answers from eval_responses.json.

9.2 Overall Results

Qwen3-VL-4B achieved an overall average score of **2.17 / 3.00 (72.3%)** across all 100 questions. Mistral-7B-Instruct-v0.3 achieved **1.50 / 3.00 (50.0%)**. Qwen leads by **0.67 points — a 45% relative advantage** on per-question accuracy.

Model	Avg Score	% of Max	Fully Correct (3)	Partial (2)	Mostly Wrong (1)	Zero (0)
Qwen3-VL-4B	2.17 / 3.00	72.3%	40 (40%)	42 (42%)	13 (13%)	5 (5%)
Mistral-7B	1.50 / 3.00	50.0%	7 (7%)	54 (54%)	21 (21%)	18 (18%)

9.3 Per-Category Breakdown

The table below shows average scores per knowledge category. Qwen leads in every single category. The largest gaps are in vLLM Flags & Configuration (+0.87) and Replicas vs Tensor Parallelism (+0.90) — areas where Mistral most frequently answered from the wrong domain context.

Category	Qs	Qwen	Mistral	Qwen Lead
KV Cache	15	2.67	2.27	+0.40
TTFT & ITL Diagnosis	15	2.07	1.40	+0.67
Batching & Scheduling	15	2.20	1.73	+0.47
GPU Memory & Quantisation	15	2.27	1.47	+0.80
vLLM Flags & Configuration	15	2.20	1.33	+0.87
Replicas vs Tensor Parallelism	10	2.10	1.20	+0.90

Chunked Prefill & Prefix Caching	15	1.67	1.00	+0.67
Overall	100	2.17	1.50	+0.67

9.4 Key Qualitative Findings

- **Qwen strongest on KV Cache (2.67/3.00).** Qwen demonstrated near-complete mastery of KV cache mechanics — memory allocation, PagedAttention, GQA, eviction, prefix caching, and bandwidth saturation. 11 of 15 questions scored 3 (fully correct).
- **Mistral's dominant failure mode: wrong domain context.** Mistral answered 18 questions with a score of 0, frequently substituting entirely wrong domains — defining ITL as 'In-Memory Table Limit in Apache Flink', TTFT as 'Time to First Text' in web performance, and GQA as a key-value store query optimisation technique. This pattern indicates Mistral-7B is not reliably grounded in vLLM-specific inference engineering knowledge.
- **Largest Qwen advantage: vLLM flags and parallelism (Section 61–85).** On vLLM Flags & Configuration, Qwen scored 2.20 vs Mistral's 1.33. Mistral repeatedly confused vLLM flags with TensorFlow or generic PyTorch parameters. On Replicas vs Tensor Parallelism, Qwen scored 2.10 vs Mistral's 1.20. Both categories are directly relevant to a production inference engineering tutor.
- **Both models weakest on Chunked Prefill & Prefix Caching (86–100).** Qwen scored 1.67 and Mistral 1.00 on this category — the most technically advanced section. Both models incorrectly stated that prefix caching does not work across different users (Q90), and neither correctly explained the online softmax / Flash Attention relationship (Q99). This is the area requiring the most improvement in both models.
- **Score distributions reveal different error profiles.** Qwen's errors skew toward partial credit (42 scores of 2) — mostly right with missing detail. Mistral's errors skew toward zero (18 completely wrong) — fundamental misunderstandings. For a production tutor, Mistral's failure mode (confidently wrong answer in a different domain) is more harmful than Qwen's (correct concept, incomplete explanation).

9.5 Implications for Deployment Recommendation

The quality evaluation resolves the open question from Section 6.4. There is no quality-vs-latency trade-off to make: Qwen3-VL-4B wins on both dimensions. The table below summarises the complete evidence base for the deployment decision.

Dimension	Qwen3-VL-4B	Mistral-7B	Winner
Serving: Composite score at 100u	67.8M	49.0M	Qwen +38%
Serving: ITL p95 stability	5.6–7.1ms (stable)	Spikes to 29.6ms at 150u	Qwen ✓
Serving: Peak throughput	4,528 chunks/s	3,960 chunks/s	Qwen +14%
Quality: Overall avg score (0–3)	2.17 / 3.00 (72%)	1.50 / 3.00 (50%)	Qwen +45%
Quality: Fully correct answers	40 / 100 (40%)	7 / 100 (7%)	Qwen ✓
Quality: Hallucination rate (0s)	5 / 100 (5%)	18 / 100 (18%)	Qwen ✓
Quality: Best category (KV Cache)	2.67 / 3.00	2.27 / 3.00	Qwen +0.40
Quality: Worst category (Chunked)	1.67 / 3.00	1.00 / 3.00	Qwen +0.67

Confirmed recommendation: Deploy Qwen3-VL-4B at 100 concurrent users per H100. This configuration delivers the highest composite serving score (67.8M), stable ITL (5.6 ms p95), and the highest domain knowledge accuracy (2.17/3.00). Qwen3-VL-4B is the production deployment choice on every measured dimension.